

GARNER FIELD, TX, USA

WMO: 722202

Lat: 29.211N

Lon: 99.744W

Elev: 942

StdP: 14.20

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12985

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.4	32.9	9.2	9.1	45.8	13.1	11.1	46.5	24.1	62.6	20.0	60.9	3.6	50	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	23.2	100.3	72.3	99.2	72.3	97.7	72.5	78.4	88.9	77.5	88.4	76.8	87.5	9.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.7	139.1	81.5	75.0	135.8	80.8	74.1	131.8	80.1	42.6	89.0	41.6	88.0	40.8	87.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.1	17.0	15.4	DB	22.3	104.5	3.8	3.2	19.6	106.8	17.4	108.7	15.2	110.5	12.5	112.8
			WB	20.5	80.6	3.7	1.4	17.8	81.6	15.7	82.4	13.6	83.2	10.9	84.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	52.6	57.3	63.9	70.8	77.2	82.9	84.3	85.1	80.2	71.7	61.2	53.4
	DBStd	13.47	8.29	9.31	8.33	7.27	6.02	3.88	3.05	3.21	4.28	7.55	8.59	8.09
	HDD50	183	67	36	7	1	0	0	0	0	0	0	16	56
	HDD65	1353	393	239	124	27	3	0	0	0	0	32	167	368
	CDD50	7524	146	241	437	625	843	988	1064	1088	906	672	351	163
	CDD65	3221	7	25	89	202	381	538	599	623	456	239	53	9
	CDH74	35951	83	254	701	1905	3786	6402	7448	8179	4688	2070	360	75
	CDH80	18512	11	59	200	772	1794	3497	4229	4792	2334	752	66	6
Wind	WSAvg	6.8	5.7	6.6	7.7	8.1	8.2	8.0	7.2	6.9	6.1	6.3	5.6	5.3
Precipitation	PrecAvg	24.6	1.1	1.3	1.1	2.3	3.1	2.7	1.9	3.0	2.8	2.8	1.3	1.1
	PrecMax	46.1	3.7	3.9	4.3	7.9	7.9	14.9	10.1	15.8	9.2	13.4	3.1	3.9
	PrecMin	13.6	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.2	0.0	0.0	0.0
	PrecStd	7.4	0.8	1.1	1.1	2.0	2.0	3.1	2.7	3.5	2.3	2.9	1.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.8	87.5	90.2	97.4	99.5	102.3	102.5	102.3	99.4	93.1	87.6	80.6
		MCWB	60.1	62.2	63.2	66.8	71.7	72.3	72.2	72.4	69.7	72.8	67.5	61.0
	2%	DB	75.7	80.9	86.5	92.6	96.8	99.6	100.1	100.2	96.8	90.2	81.4	75.5
		MCWB	59.8	60.0	65.5	67.8	72.5	72.4	72.5	72.6	71.8	71.1	65.3	61.2
	5%	DB	72.3	76.8	81.3	88.2	93.2	97.6	98.6	99.1	94.1	88.0	78.3	72.5
		MCWB	57.9	60.4	63.6	68.3	72.0	72.5	72.6	72.7	72.3	70.5	64.4	60.2
	10%	DB	68.2	72.7	77.7	84.3	89.8	95.1	96.4	97.5	91.2	85.2	74.8	68.4
		MCWB	57.3	59.7	62.7	66.9	72.1	73.1	73.1	73.1	71.5	69.9	63.4	59.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.4	68.8	71.2	75.0	78.3	79.8	79.3	78.6	78.5	78.4	72.9	69.4
		MCDB	72.1	75.3	79.0	86.2	89.2	89.7	89.7	89.8	88.7	85.4	78.0	72.8
	2%	WB	66.0	67.1	69.4	73.4	77.0	78.3	77.9	77.6	77.2	75.8	70.8	67.5
		MCDB	71.3	72.5	76.5	83.4	87.4	89.0	88.9	89.1	87.1	82.8	75.8	71.6
	5%	WB	62.6	65.6	68.3	72.2	75.8	77.2	77.0	76.9	76.2	74.3	68.8	65.6
		MCDB	68.1	70.8	75.0	80.9	85.7	88.2	88.1	88.4	85.3	81.2	74.0	68.9
	10%	WB	58.5	63.4	67.0	71.0	74.7	76.2	76.1	76.0	75.3	73.1	67.1	61.7
		MCDB	64.3	69.4	73.4	78.9	84.2	86.8	87.0	87.5	83.8	79.8	72.6	66.3
Mean Daily Temperature Range	5% DB	MDBR	23.7	23.4	23.3	24.2	21.8	21.9	22.6	23.2	20.6	22.6	22.3	22.9
		MCDBR	30.2	29.2	26.5	28.3	26.8	26.0	26.3	25.5	24.4	24.0	24.5	28.9
		MCWBR	17.0	15.5	11.9	12.0	9.3	6.7	6.6	6.9	7.5	9.6	13.1	16.7
	5% WB	MCDBR	21.4	20.5	20.4	21.1	20.4	20.8	22.1	22.9	18.9	18.5	19.1	19.2
		MCWBR	15.2	14.1	11.3	9.9	8.1	7.0	6.6	6.6	6.6	8.8	12.2	15.0
Clear-Sky Solar Irradiance	taub	0.324	0.330	0.351	0.383	0.411	0.410	0.412	0.404	0.394	0.365	0.343	0.321	
	taud	2.497	2.478	2.440	2.373	2.345	2.402	2.422	2.437	2.408	2.481	2.476	2.511	
	Ebn at Noon	291	298	297	289	279	278	277	279	279	282	281	286	
	Edh at Noon	29	32	35	39	40	38	37	36	36	31	29	27	
All-Sky Solar Radiation	RadAvg	995	1198	1490	1756	1847	2130	2133	2045	1666	1369	1053	893	
	RadStd	117	194	156	178	206	159	181	113	154	191	133	125	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+241	+187

Nomenclature: See separate page